#1)

(1) $A-\lambda I = \begin{pmatrix} 4-\lambda & 2 \\ 2 & 1-\lambda \end{pmatrix} \Rightarrow$ characteristic polynomial : $\det(A-\lambda I)$

$= (4-\lambda)(1-\lambda)-4$

$= \lambda^2 - 5\lambda = \lambda(\lambda-5)$

$\Rightarrow \lambda_1 = 0,\ \lambda_2 = 5$

$\Rightarrow (A-\lambda I)x = 0$

case 1) $\lambda_1 = 0$

$$\begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \qquad \begin{matrix} 4x_1+2x_2=0 \\ 2x_1+x_2=0 \end{matrix} \Rightarrow x_2 = -2x_1$$

$$\Rightarrow v_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

$$\Rightarrow u_1 = \begin{pmatrix} \frac{1}{\sqrt{5}} \\ -\frac{2}{\sqrt{5}} \end{pmatrix}$$

case 2) $\lambda_2 = 5$

$$\begin{pmatrix} -1 & 2 \\ 2 & -4 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \qquad \begin{matrix} -x_1+2x_2=0 \\ 2x_1-4x_2=0 \end{matrix} \Rightarrow x_1 = 2x_2$$

$$\Rightarrow v_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$\Rightarrow u_2 = \begin{pmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{pmatrix}$$

답: $\begin{cases} \lambda_1 = 0,\ u_1 = \begin{pmatrix} \frac{1}{\sqrt{5}} \\ -\frac{2}{\sqrt{5}} \end{pmatrix} \\ \lambda_2 = 5,\ u_2 = \begin{pmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{pmatrix} \end{cases}$

(2) $P = [v_1 \ v_2] = \frac{1}{\sqrt{5}}\begin{pmatrix} 1 & 2 \\ -2 & 1 \end{pmatrix}$, $D = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 5 \end{pmatrix}$

And since P is orthogonal, $PP^T = I \Rightarrow P^T = P^{-1}$

Thus $A = \begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix} = \frac{1}{\sqrt{5}}\begin{pmatrix} 1 & 2 \\ -2 & 1 \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 0 & 5 \end{pmatrix}\frac{1}{\sqrt{5}}\begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix}$

$= PDP^T$ (eigenvalue decomposition) //

(3) Since $\lambda_1 = 0, \lambda_2 = 5$, A is p.s.d. (all eigenvalue ≥ 0)

(4) Let $\begin{cases} v_1, v_2 : \text{eigenvectors of } A. \\ \lambda_1, \lambda_2 : \text{corresponding eigenvalues of } A. \end{cases}$

(i.e. $Av_1 = \lambda v_1$, $Av_2 = \lambda v_2$)

Since A is not symmetric, $A \neq A^T$. Thus we can't ensure that $Av_k = A^T v_k$ (v_k : eigenvector)

$\therefore v_1^T A v_2 = v_1^T(\lambda_2 v_2) = \lambda_2 v_1^T v_2$ and

$(Av_1)^T v_2 = (\lambda v_1)^T v_2 = \lambda v_1^T v_2$

Since $A \neq A^T$, we can't induce $\lambda_2 v_1^T v_2 = \lambda_1 v_1^T v_2$

(i.e. $(\lambda_2 - \lambda_1) v_1^T v_2 = 0$) so we can't reach to the conclusion that $v_1^T v_2 = 0$. ∎

실수 포인 : 잘 모르겠습니다.

#1-2

(1) $B^TB = \begin{pmatrix} 2 & 1 & 2 \\ 2 & -1 & 2 \end{pmatrix} \begin{pmatrix} 2 & 2 \\ 1 & -1 \\ 2 & 2 \end{pmatrix} = \begin{pmatrix} 9 & 7 \\ 7 & 9 \end{pmatrix}$

$BB^T = \begin{pmatrix} 2 & 2 \\ 1 & -1 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} 2 & 1 & 2 \\ 2 & -1 & 2 \end{pmatrix} = \begin{pmatrix} 8 & 0 & 8 \\ 0 & 2 & 0 \\ 8 & 0 & 8 \end{pmatrix}$

(2)

$B^TB - \lambda I = \begin{pmatrix} 9-\lambda & 7 \\ 7 & 9-\lambda \end{pmatrix} \Rightarrow \det(B^TB - \lambda I) = (9-\lambda)^2 - 49$

$= \lambda^2 - 18\lambda + 32$

$\Rightarrow$ eigenvalues : $\lambda_1 = 2,\ \lambda_2 = 16.$

case1) $\lambda_1 = 2$

$\begin{pmatrix} 7 & 7 \\ 7 & 7 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \rightarrow x_1 + x_2 = 0 \quad v_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \Rightarrow u_1 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix}$

case2) $\lambda_2 = 16.$

$\begin{pmatrix} -7 & 7 \\ 7 & -7 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \rightarrow -x_1 + x_2 = 0 \quad v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \Rightarrow u_2 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$

Thus $V = (u_1\ u_2) = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$

$\sigma_1 = \sqrt{16} = 4$

$\sigma_2 = \sqrt{2} = \sqrt{2}$

(고쳐본: $V = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}$

(3) $BB^T = \begin{pmatrix} 8 & 0 & 8 \\ 0 & 2 & 0 \\ 8 & 0 & 8 \end{pmatrix} \Rightarrow (BB^T - \lambda I) = \begin{pmatrix} 8-\lambda & 0 & 8 \\ 0 & 2-\lambda & 0 \\ 8 & 0 & 8-\lambda \end{pmatrix}$

$$\det(BB^T - \lambda I) = (8-\lambda)(2-\lambda)(8-\lambda) - 64(2-\lambda)$$
$$= (2-\lambda)(64 - 16\lambda + \lambda^2 - 64)$$
$$= (2-\lambda)(\lambda^2 - 16\lambda) = (2-\lambda)(\lambda)(\lambda - 16)$$

$\lambda_1 = 16$, $\lambda_2 = 2$, $\lambda_3 = 0$

case① $\lambda_1 = 16$)

$$\begin{pmatrix} -8 & 0 & 8 \\ 0 & -14 & 0 \\ 8 & 0 & -8 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow x_1 = x_3,\ x_2 = 0 \Rightarrow v_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

$$\Rightarrow u_1 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

case② $\lambda_2 = 2$)

$$\begin{pmatrix} 6 & 0 & 8 \\ 0 & 0 & 0 \\ 8 & 0 & 6 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow \begin{array}{l} -3x_1 = 4x_2 \\ -4x_1 = -3x_2 \end{array} \Rightarrow u_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

case③ $\lambda_3 = 0$)

$$\begin{pmatrix} 8 & 0 & 8 \\ 0 & 2 & 0 \\ 8 & 0 & 8 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow \begin{array}{l} x_1 = -x_3 \\ x_2 = 0 \end{array} \Rightarrow u_3 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ -\frac{1}{\sqrt{2}} \end{pmatrix}$$

$$U = (u_1\ u_2\ v_3) = \begin{pmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \end{pmatrix}$$

$$B=\begin{pmatrix}2&2\\1&-1\\2&2\end{pmatrix}=\begin{pmatrix}\frac{1}{\sqrt2}&0&\frac{1}{\sqrt2}\\0&1&0\\\frac{1}{\sqrt2}&0&-\frac{1}{\sqrt2}\end{pmatrix}\begin{pmatrix}4&0\\0&\sqrt2\\0&0\end{pmatrix}\begin{pmatrix}\frac{1}{\sqrt2}&\frac{1}{\sqrt2}\\\frac{1}{\sqrt2}&-\frac{1}{\sqrt2}\end{pmatrix}$$

고유값의 크기 순서 재배치함

#1-3

(1) $\Sigma-\lambda I=\begin{pmatrix}5-\lambda&3\\3&5-\lambda\end{pmatrix}\Rightarrow \det(\Sigma-\lambda I)=(5-\lambda)^2-9$

$=\lambda^2-10\lambda+16$

$=(\lambda-2)(\lambda-8)$

$\therefore \lambda_1=8,\ \lambda_2=2.$

Case1) $\lambda_1=8.$

$\begin{pmatrix}-3&3\\3&-3\end{pmatrix}\begin{pmatrix}x_1\\x_2\end{pmatrix}=\begin{pmatrix}0\\0\end{pmatrix}\Rightarrow x_1=x_2\Rightarrow u_1=\begin{pmatrix}\frac{1}{\sqrt2}\\\frac{1}{\sqrt2}\end{pmatrix}$

Case2) $\lambda_2=2$

$\begin{pmatrix}3&3\\3&3\end{pmatrix}\begin{pmatrix}x_1\\x_2\end{pmatrix}=\begin{pmatrix}0\\0\end{pmatrix}\Rightarrow x_1=-x_2\Rightarrow u_2=\begin{pmatrix}\frac{1}{\sqrt2}\\-\frac{1}{\sqrt2}\end{pmatrix}$

(2) By sizes of eigenvalues, $\lambda_1=8,\ \lambda_2=2.$

$PC1=\begin{pmatrix}\frac{1}{\sqrt2}\\\frac{1}{\sqrt2}\end{pmatrix}\quad PC2=\begin{pmatrix}\frac{1}{\sqrt2}\\-\frac{1}{\sqrt2}\end{pmatrix}$

(3) PC1의 분산비율: $\frac{8}{8+2}=80\%$ / PC2 〃 : $\frac{2}{8+2}=20\%.$

(4) 20% 이다. PC2의 분산비율이 20%이므로.

#2 (1) $p(A) = \frac{8}{16}$ $p(B) = \frac{4}{16}$ $p(C) = \frac{4}{16}$

(2) $H(X) = -\sum_{x} p(x) \log_2 p(x)$

$= -\left(\frac{8}{16}\cdot(-1) + \frac{4}{16}\cdot(-2) + \frac{4}{16}\cdot(-2)\right) = \frac{24}{16} = 1.5$

(3) 최대 entropy : $p(x_i) = \frac{1}{3}$ $\forall i \in [3]$ 시 발생

$H_{max} = -\frac{1}{3}\log_2\left(\frac{1}{3}\right) \times 3 = \log_2 3$

$H_{max} = \log_2 3 > 1.5 = H(X)$이므로 현재 data entropy보다 큼 ⇒ 데이터 불확실성이 더 높다.

#2-2 (1) $D_{KL}(P||Q) = 0.5\log_2 2 + 0.3\log_2 0.6 + 0.2\log_2 0.8$

$\neq$

$D_{KL}(Q||P) = 0.25\log_2 \frac{1}{2} + 0.5\log_2\left(\frac{5}{3}\right) + 0.25\log_2 1.25$

→ not a metric.

(2) KL divergence : 두 확률분포 간의 거리 역할.

값이 0이다 ⟺ 두 분포가 동일하다. (i.e. $P = Q$)

#3-1

(1) $\bar{x} = \frac{1+2+3+4+5}{5} = 3$, $\bar{y} = \frac{2+3+5+6+9}{5} = \frac{25}{5} = 5$.

$S_{xy} = \sum(x_i-\bar{x})(y_i-\bar{y}) = \sum x_i y_i - n\bar{x}\bar{y}$

$= 2+6+15+24+45 - 5\times3\times5 = 92-75 = 17$.

$S_{xx} = \sum(x_i-\bar{x})^2 = \sum x_i^2 - n\bar{x}^2 = \frac{5\times6\times11}{6} - 5\times9 = 55-45 = 10$

(2) $\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = 1.7$, $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x} = 5 - 1.7\times3 = -0.1$

$\therefore \hat{y} = -0.1 + 1.7x$.

(3)

y_i	$\hat{y}_i$	e_i	e_i^2
2	1.6	0.4	0.16
3	3.3	-0.3	0.09
5	5.0	0	0
6	6.7	-0.7	0.49
9	8.4	0.6	0.36

$\Rightarrow SSE = 0.25 + 0.85 = 1.10$.

$SST = S_{yy} = \sum y_i^2 - n\bar{y}^2$

$= (4+9+25+36+81) - 5\cdot25$

$= 30$.

$\Rightarrow R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST} = \frac{30-1.1}{30} = \frac{28.9}{30}$

(4) $R^2 \approx 1$ 이므로 회귀선을 매우 잘 설명한다.

#3-2

(1) $VIF_{x_1} = \frac{1}{1-R_1^2} = 10$.

(2) $VIF_{x_1} = 10$ 이므로 매우 강한 다중공선성 가정

$\Rightarrow X_1$의 제거를 고려해야 한다.

#4 (1) $d(q, A_1) = \sqrt{9+9} = 3\sqrt{2}$

$d(q, A_2) = \sqrt{0.5^2+0} = 0.5$

$d(q, B_1) = \sqrt{1+1} = \sqrt{2}$

$d(q, B_2) = \sqrt{1+1} = \sqrt{2}$

$d(q, B_3) = \sqrt{2^2+0} = 2$

(2) nearest 3 neighbors: A_2, B_1, B_2 ⇒ B가 더 많음 ⇒ q의 class는 B이다.

(3) $w_{A_2} = \dfrac{1}{d(X_q, X_{A_2})} = \dfrac{1}{0.5} = 2$

$w_{B_1} = \dfrac{1}{d(X_q, X_{B_1})} = \dfrac{1}{\sqrt{2}} = w_{B_2}$

class A의 가중치 합 $= 2$ > class B의 가중치 합 $= 2 \times \frac{1}{\sqrt{2}} = \sqrt{2}$.

∴ q의 class는 A이다.

(4) 결과가 다르다. $K=3$인 두 결과에서 기존의 KNN은 단순히 다수결로 정하나, WKNN의 경우 개수뿐 아니라 가중치의 합을 고려 ⇒ A가 훨씬 가깝기 때문에 가중값이 큰 A에 의해 class A가 된다

#5

(1) $d(p_1, c_1) = 0,\ d(p_1, c_2) > 0 \Rightarrow p_1 \to c_1$

$d(p_2, c_1) = 2,\ d(p_2, c_2) = \sqrt{36+49} = \sqrt{85} \Rightarrow p_2 \to c_1$

$d(p_3, c_1) = \sqrt{2},\ d(p_3, c_2) = \sqrt{25+36} > \sqrt{2} \Rightarrow p_3 \to c_1$

$d(p_4, c_1) = \sqrt{36+25},\ d(p_4, c_2) = 0 \Rightarrow p_4 \to c_2$

$d(p_5, c_1) = \sqrt{49+25},\ d(p_5, c_2) = 1 \Rightarrow p_5 \to c_2$

$d(p_6, c_1) = \sqrt{36+36},\ d(p_6, c_2) = 1 \Rightarrow p_6 \to c_2$

$\therefore\ p_1, p_2, p_3 \to c_1$

$p_4, p_5, p_6 \to c_2$

(2) $\overline{c_1} = \left(\frac{2+2+3}{3}, \frac{3+1+2}{3}\right) = \left(\frac{7}{3}, 2\right),\ \overline{c_2} = \left(\frac{8+9+8}{3}, \frac{8+8+9}{3}\right) = \left(\frac{25}{3}, \frac{25}{3}\right)$

(3) $d(p_1, \overline{c_1}) = \sqrt{\left(\frac{1}{3}\right)^2 + 1} = \frac{\sqrt{10}}{3},\ d(p_1, \overline{c_2}) = \sqrt{\left(\frac{19}{3}\right)^2 + \left(\frac{16}{3}\right)^2} \Rightarrow p_1 \to \overline{c_1}$

$d(p_2, \overline{c_1}) = \sqrt{\left(\frac{1}{3}\right)^2 + 1} = \frac{\sqrt{10}}{3},\ d(p_2, \overline{c_2}) = \sqrt{\left(\frac{19}{3}\right)^2 + \left(\frac{22}{3}\right)^2} \Rightarrow p_2 \to \overline{c_1}$

$d(p_3, \overline{c_1}) = \sqrt{\left(\frac{2}{3}\right)^2 + 0} = \frac{2}{3},\ d(p_3, \overline{c_2}) = \sqrt{\left(\frac{16}{3}\right)^2 + \left(\frac{19}{3}\right)^2} \Rightarrow p_3 \to \overline{c_1}$

$d(p_4, \overline{c_1}) = \sqrt{\left(\frac{17}{3}\right)^2 + 36},\ d(p_4, \overline{c_2}) = \sqrt{\left(\frac{1}{3}\right)^2 + \left(\frac{1}{3}\right)^2} \Rightarrow p_4 \to \overline{c_2}$

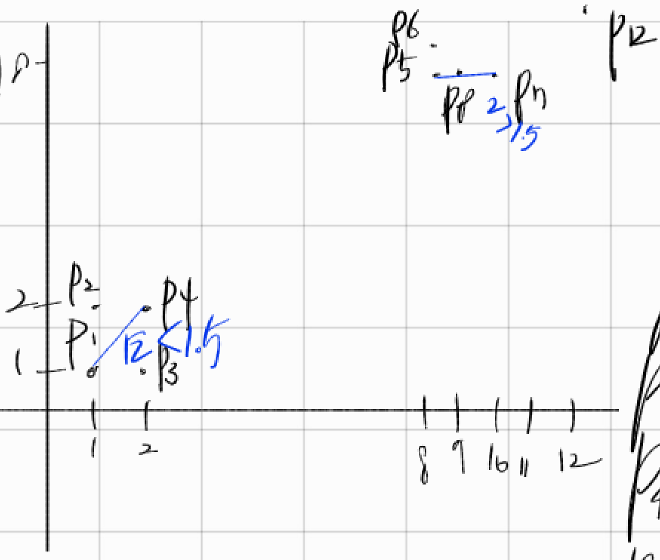
$d(p_5, \overline{c_1}) = \sqrt{\left(\frac{20}{3}\right)^2 + 36},\ d(p_5, \overline{c_2}) = \sqrt{\left(\frac{2}{3}\right)^2 + \left(\frac{1}{3}\right)^2} \Rightarrow p_5 \to \overline{c_2}$

$d(p_6, \overline{c_1}) = \sqrt{\left(\frac{17}{3}\right)^2 + 49},\ d(p_6, \overline{c_2}) = \sqrt{\left(\frac{1}{3}\right)^2 + \left(\frac{1}{3}\right)^2} \Rightarrow p_6 \to \overline{c_2}$

$\Rightarrow$ [illegible]

$\Rightarrow$ K-means algorithm 종료.

#5-2



(1)

$p_1 \Rightarrow p_1, p_2, p_3, p_4$ (4개)

$p_2 \Rightarrow p_1, p_2, p_3, p_4$ (4개)

$p_3 \Rightarrow p_1, p_2, p_3, p_4$ (4개)

$p_4 \Rightarrow p_1, p_2, p_3, p_4$ (4개)

$p_5 \Rightarrow p_5, p_6, p_8$ (3개)

$p_6 \Rightarrow p_5, p_6, p_8$ (3개)

$p_7 \Rightarrow p_7, p_8$ (2개)

$p_8 \Rightarrow p_5, p_6, p_7, p_8$ (4개)

$p_9 \Rightarrow p_9$ (1개)

(2) Since minPts = 3,

Core pt = $(p_1, p_2, p_3, p_4, p_5, p_6, p_8)$

Border pt = (p_7)

Noise pt = (p_9)

(3) 군집. $A = (p_1, p_2, p_3, p_4)$

$B = (p_5, p_6, p_7, p_8)$

p_9 : 어느 군집에도 X.

By DBSCAN algorithm,

$p_1 \rightarrow p_2, p_3, p_4$

$p_5 \rightarrow p_6 \rightarrow p_8 \rightarrow p_7$.

(계속 변화)

K-means : 초기가 어디에 찍힐 지에 centroid이 변경 ⇒ centroid 결정.

DBSCAN : 초기 ε, minPts값 설정 ⇒ 알고리즘 ⇒ 자동 군집 형성

(데이터의 밀도, 분포의 영역 자동) ⇓

Centroid 불필요